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ARTIFICIAL INTELLIGENCE

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Assignment Documentation

Project Title: Used Car Price Prediction using Supervised Machine Learning

Programme: RDS Y2S2

Tutorial Group: 2

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1.0 Introduction

1.1 Problem Background

The global used car market is growing rapidly, valuing over $1.5 trillion in 2023. There is indicating a continued expansion due to the increased demand for affordable vehicles (Statista, 2023). However, pricing the used cars accurately still remains a significant challenge in today's environment. It is because the price of used cars will influence by various of factors, such as:

- **Vehicle attributes** :
    - Age
        - The older cars will depreciate in value generally.
    - Mileage
        - Typically, the higher mileage, the lower the price due to increased wear and tear.
    - Brand
        - Some of the brands such as Toyota or Honda will hold the value better.
    - Engine capacity
        - A well-maintained engine will increase the value.

- **Market trends** :
    - Supply and demand
        - The popular models in short supply will increase the prices of cars.
    - Fuel prices
        - If the fuel costs are increased, the demand for economical cars will increase. Then affect the price for economical cars.
    - Economic conditions
        - The recessions may lower the demand for luxury cars.
    - New car availability
        - If new cars are scarce, the price of used cars will rise.

- **Condition and history** :
    - Accident records

        - The cars with the accident records such as serious traffic accidents or flood caused water to seep into the car will have a lower value.
    - service history
        - A full service record increases buyer confidence and price.
    - ownership history
        - Fewer previous owners usually mean a higher price.
    - Body & Paint Condition
        - If the car has dents, rust and scratches reduce the price.

Based on those factors that mentioned above, the traditional valuation methods such as Kelley Blue Book (KBB) or dealership appraisals are created as a guideline for used car pricing. Those methods often rely on generalized formulas or subjective assessments, which may not account for dynamic market fluctuations or hidden defects. This leads to inconsistency, undervaluation, or overpricing, creating distrust among buyers and sellers.

This will lead to the overpaying by the buyers. It means that the buyers may waste money to buy a used car with a price that is not reasonable. For example the buyers that lack research before buying a used car. Not only that, the buyers may fall for a “too good to be true” scams. For example, the scammers list the cars below the market value to lure buyers. If the buyers lack previous knowledge, they will make the decision by the pressure from the sellers.

Besides, the sellers will also struggle to sell the cars. They may face the problem that overpriced cars sit unsold for months. The seller will lose the potential buyers or may have to sell at a much lower price later.

To solve these problems, this project uses the "Used Car Price Prediction Dataset" (Najib, 2024) to create a used car price prediction model. This dataset is a comprehensive collection that contain 4,009 listings with detailed features like:

- Brand and model
- Model year
- Mileage
- Fuel type
- Engine type

- Transmission
- Exterior and interior colors
- Accident history
- Clean title

1.2 Objectives/Aims

With those features provided by the dataset, this project uses supervised machine learning (ML) to train a model. This project aims to develop a data-driven pricing model that improves fairness, accuracy, transparency, fraud prevention and efficiency in the used cars market. So that the market has an accurate measurement for the used car prices based on the car conditions. Before training this car price prediction model, practical usability and insightful analysis must be ensured.

1. **Develop a Robust Prediction Model**
    - Train and compare multiple supervised learning algorithms, including linear regression, random forest and XGBoost to predict the prices of used cars based on key features such as mileage, age, brand, and condition. The goal of this model is to achieve a high accuracy used car prediction with minimal error using evaluation metrics like Mean Absolute Error (MAE) and R-squared (R^2) to measure the model's performance.
2. **Identify Key Price Influencers**
    - Analyze feature importance to determine which factors such as low mileage or popular car brands have the strongest impact on a car's resale value. These insights can assist buyers, sellers, and dealers in making more informed decisions.
3. **Create a User-Friendly Prototype**
    - Develop a simple, interactive tool such as a Streamlit web app or Python script where users can input car details and receive an instant price estimate. This demonstrates the model's real-world usability and accessibility.
4. **Compare Model Performance**
    - Compare each algorithm's strengths and weaknesses in terms of accuracy, speed, and interpretability. This comparison will help identify the most suitable method for this specific prediction task.
5. **Ensure Reproducibility and Scalability**
    - Document the entire workflow—from data cleaning to model deployment—so others can replicate or expand the project. The solution should be adaptable to larger datasets or regional markets.

1.3 Motivation

The used car market is a massive global industry, yet pricing remains inconsistent and often opaque, leading to distrust between buyers and sellers. This project has significant commercial potential and social impact by introducing a data-driven approach to price prediction.

Commercial Potential

- **For Car Dealerships and Online Platforms**

  Car dealerships and websites like CarMax, AutoTrader, or OLX could use this model to offer instant, fair price estimates. It would save time, cut down manual appraisal costs, and help set dynamic prices based on real-time market trends to stay competitive and maximize profits.

- **For Startups and Fintech Companies**

  A startup could turn this into a simple app where anyone can check a car's value before buying or selling. Companies like banks and insurance could also use the model to estimate car values for loans or insurance claims.

- **For Manufacturers and Fleet Managers**

  Car makers could use this information to figure out which features (like fuel efficiency or strong brand reputation) help cars keep their value longer, guiding future car designs. Fleet managers could predict resale values for lease returns or for selling off rental cars more accurately.

Social Impact

- **Empowering Consumers**

  By providing fair and unbiased price estimates, buyers and sellers can have better information and lesser reasons to argue during negotiations. It would also builds trust and make the used car market more transparent.

- **Promoting Sustainable**

  With better pricing, it makes it easier for people to buy good used cars instead of always choosing the new ones. This reduces waste and helps lower carbon emissions by extending the life of vehicles.

- **Supporting Economic Accessibility**

  Clear and reliable pricing would also help families that have lower incomes to find affordable cars, making the transportation more accessible. It also gives small dealerships and independent sellers a better chance to compete with bigger companies.

This project shows how machine learning can solve real-world problems outside of tech fields. It also provides a strong starting point for future research into AI applications in the automotive industry.

By combining data science with real-world business and social needs, this project aims not just to predict prices but to help create a fairer, greener, and more open used car market for everyone.

2.0 Research Background

2.1 Background of the applications

Setting the right price for a used car is really important for both buyers and sellers, especially now with so many transactions happening in the online market. In the past, people usually had to rely on car experts to estimate a car's value. But this traditional way of pricing was often slow, inconsistent, and sometimes even biased (Sivakumar & Anitha, 2021). Thanks to the rise of big data and machine learning, it's now possible to predict car prices much faster and more accurately using automated systems.

Many studies have used machine learning algorithms like Linear Regression, Decision Trees, Random Forest, and Support Vector Machines (SVM) to predict used car prices (Nair & Sreekumar, 2021). However, basic models like Linear Regression don't always capture the complicated relationships in the data , such as how a car's brand, mileage, or age affects its value. More powerful models like Random Forest and Gradient Boosting have shown better results because they can handle these complex patterns (Palaniappan & Awang, 2018).

Recent research also showed that properly cleaning the data and creating new, meaningful features (like calculating the car's age from its manufacture year) can make a big difference in prediction accuracy (Aravindhan et al., 2020). Even so, there are still challenges. Many models don't work equally well across different types of cars (luxury vs economy) or different markets. Some models are also difficult for users to understand, which can make it hard for people to trust their predictions.

This project aims to build a reliable and easy-to-understand system for predicting used car prices by trying out different machine learning algorithms, improving the data features, and making sure the model's results are not only accurate but also easy to explain.

2.2 Analysis of selected tool with any other relevant tools.

Tools comparison	**Remark**	**Jupyter Notebook (Selected)**	**Google Colab**	**RStudio (Posit)**
Type of license and open source license	State all types of license	Modified BSD (Permissive)	Proprietary (Jupyter-based)	AGPLv3 (Open Source), Proprietary (Commercial)
Year founded	When is this tool being introduced?	2014	2017	2008
Founding company	Owner	Project Jupyter (Community)	Google	RStudio, Inc. (now Posit, PBC)
License Pricing	Compare the prices if the license is used for development and business/commercialization	Free	Free (Limited), Paid ($9.99–$49.99/month)	Free (Open Source), Paid (~$5,000+/year)
Supported features	What features does it offer?	Multi-language, interactive coding, visualization	Cloud-based, GPU/TPU, Python-focused	R-focused, visualization, ML packages, Shiny
Common applications	Data science, ML, education	Machine learning projects, academic and industry prototyping	ML, deep learning, education	Statistical analysis, data science, BI
Customer support	How the customer support is given, e.g. proprietary, online community, etc.	Community-driven	Community + Limited Proprietary	Community + Proprietary (Commercial)
Limitations	The drawbacks of the software	Local setup, scalability, version control	Resource limits, Python-only, privacy	R-centric, costly commercial licenses, performance

2.3 Justify why the selected tool is suitable

Jupyter Notebook was chosen for this project because it fits perfectly with the needs of a supervised machine learning workflow. It allows us to handle everything from loading and cleaning data to training models and visualizing results, all in one place. It also works seamlessly with popular Python libraries like Scikit-learn, Pandas, NumPy, and Matplotlib, which are essential for building and evaluating predictive models.

One of the biggest advantages of Jupyter is its interactive environment. It lets us write and run code in smaller pieces (cells), so we can check results step-by-step, make quick changes, and test different machine learning algorithms without restarting the whole project. This is especially useful when experimenting with different models, features, or hyperparameters.

Jupyter also makes it easy to create clear, professional visualizations like graphs showing feature importance or model performance which helps when analyzing and explaining results. Plus, it supports other programming languages if needed, making it flexible for future projects.

Compared to alternatives like Google Colab or RStudio, Jupyter offers more control. It's free, open-source, and runs locally, so we don't have to worry about internet limits or losing our work due to server timeouts. Unlike RStudio, which focuses more on R programming, Jupyter is perfect for Python-based machine learning, which is what this project needs.

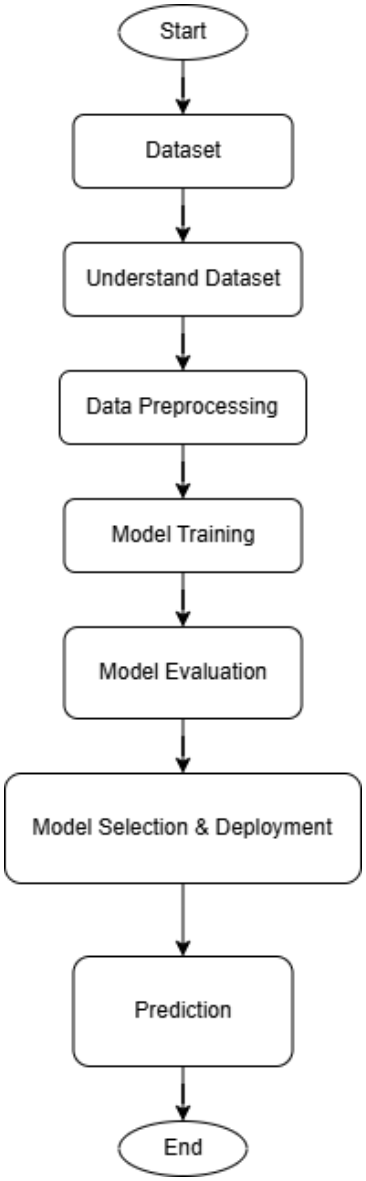
Even though Jupyter can be a bit resource-heavy for very large datasets, this isn't a problem for our project, which deals with a manageable dataset size. Also, if needed in the future, the work done in Jupyter can be expanded into cloud environments or deployed through other platforms like Flask or FastAPI.

Overall, Jupyter Notebook is a practical, cost-effective, and user-friendly choice that supports fast development, easy experimentation, and clear communication of results and are crucial for building a successful used car price prediction model.

3.0 Methodology

3.1 System flowchart/activity diagram

This model development process was initiated with the objective of predicting the price category of used cars. It has three categories, Low, Medium and High based on their key attributes. This involved several stages including understanding, preprocessing, model training, evaluation and final deployment.



Understand Dataset

Obtain the dataset (*used_cars.csv*) containing historical data of used cars. Each record included features such as :

- Brand (eg. Ford, BMW, Honda)
- Model Year
- Mileage (miles driven)
- Transmission Type (eg,. Automatic, Manual)
- Fuel Type (eg. Gasoline, Hybrid)
- Clean Title Status
- Accident History
- Listed Price

The target variable for prediction was the **Price**, which was later transformed into price categories

Data Preprocessing

Data Cleaning

- **Handling Missing Values**
    - Missing values in categorical fields (e.g., fuel_type, accident, clean_title) were filled with appropriate defaults (e.g., "Gasoline" or "No").
- **Handling Duplicates**
    - Duplicate records were checked and removed if found.
- **Handling Inconsistent Data**
    - String symbols like $ and mi. were removed from price and mileage fields. Converted text-based numbers into float format for analysis.

Feature Engineering

- Categorical Encoding
  - Categorical features (e.g., brand, fuel type, transmission) were encoded using label encoding.
  - Rare transmission types were grouped under "Other" to simplify categories.
- Normalization
  - Numerical features were normalized using StandardScaler to ensure uniform contribution across features.
- Target Variable Creation
  - The continuous price variable was binned into **THREE** categories:
    - Low (0 - 15,000)
    - Medium (15,001 - 30,000)
    - High (30,001 and above)

Splitting Dataset

- The dataset was split into:

  - **80% Training Data**: for model learning
  - **20% Testing Data**: for model evaluation

  The split was stratified to maintain a balanced distribution of the price categories.

Model Training

Model Selection

- **FOUR** supervised machine learning algorithms were selected:
  - Logistic Regression
  - K-Nearest Neighbors (KNN)
  - Support Vector Machine (SVM)
  - XGBoost Classifier

  Each model was trained using the training set. Scaled features values were used for training to improve performance. We will explain more about the model in part 3.3

Model Evaluation

Evaluation Metrics

- Model performance was evaluated using the following metrics :
    - **Accuracy** : Proportion of correct prediction
    - **Precision** : Proportion of correct positive prediction
    - **Recall** : Ability to find all positive instance
    - **F1 Score** : Harmonic mean of precision and recall

Model Comparison

- Result from all models were compared side-by-side
- A performance table and bar chart were created for visualization
- Confusion matrices were also generated to analyze prediction distribution

Model Selection and Deployment

Best Model

- XGBoost Classifier was selected as the best model based on achieving the highest accuracy, precision, recall and F1-score
- XGBoost demonstrated better handling of complex pattern compared to other models

Model Deployment

- The final trained XGBoost model was saved as a .pkl file along with the scaler
- A Streamlit application was developed to provide a simple and interactive user interface for live predictions

Prediction / Output

- Users can input new or unseen car information (e.g., brand, year, mileage, fuel type, transmission, clean title).
- The model predicts the price category (Low, Medium, High)
- The result is displayed clearly on the Streamlit interface.

3.2 Description of dataset

Dataset Source:

The dataset, named used_cars.csv is from the Kaggle website. It consists of data collected from a variety of listings of used cars. It likely originates from an automotive marketplace or a publicly available dataset repository, intended for supervised machine learning tasks such as price prediction for used vehicles.

Dataset Structure:

The dataset contains **4,009 records** with **12 columns** detailing various attributes of used cars.

Column Name	Data Type	Description	Example
brand	Object	Manufacturer of the car	Ford, Hyundai
model	Object	Specific model variant of the car	RX 350, Palisade SEL
model_year	Integer	Year the car model was manufactured	2013, 2021
milage	Object	Distance driven by the car (in miles)	"51,000 mi."
fuel_type	Object	Type of fuel used	Gasoline, Hybrid
engine	Object	Engine specification details	"3.8L V6 24V GDI DOHC"
transmission	Object	Type of car transmission system	"8-Speed Automatic"
ext_col	Object	Exterior color of the car	Blue, Black
int_col	Object	Interior color of the car	Gray, Black
accident	Object	History of accidents or damage reports	"At least 1 accident", "None reported"
clean_title	Object	Indicator whether the car has a clean title or not	Yes, NaN
price	Object	Selling price of the used car (USD)	"$10,300", "$34,999"

3.3 Description of algorithm(s)

In this project, we selected **FOUR** supervised learning algorithms and applied to predict the used car price category (Low, Medium, High). Each algorithm offers unique strengths for different types of data patterns.

Logistic Regression

Logistic Regression is a supervised machine learning algorithm designed for classification problems. Thus, it does not predict continuous values like (such as actual price like : $500). Instead, it predicts the probability that a given input belongs to a particular class or category. It uses a mathematical function called sigmoid function (S shaped curve) to convert any real-value input into a probability between 0 and 1.

In our assignment, logistic regression was employed as one of the classification algorithms to predict the price class of a used car (Low / Medium / High) based on several car attributes such as brand, model year, mileage, fuel type, transmission, and clean title status. It is simple and quick to implement.

First of all, we input the features (brand, model year, mileage, fuel type, transmission, clean title) that were first encoded (categorical to numerical) and normalized using StandardScaler. Then, the logistic regression was trained on the training dataset (80%), where the model learned how different features affect the likelihood of a car falling into the "Low", "Medium", or "High" price category. After that, the performance was evaluated based on Accuracy, Precision, Recall and F1-Score. We served Logistic Regression as a baseline model to compare against more complex models like KNN, SVM and XGBoost.

K - Nearest Neighbors algorithm (KNN)

KNN is a supervised machine learning algorithm that makes predictions based on the similarity of data points. It does not explicitly "train" a model. Instead, it memorizes the training dataset and makes decisions at prediction time. When a new data point (new car information) needs to be classified, KNN will first calculate the distance (commonly Euclidean distance) between the new point and every point in the training data. Then, find the K nearest neighbors (e.g., the 5 closest points). After that, it will assign the class that is most common among these neighbors.

In our assignment, before we applied KNN, we encoded all the categorical features and standardize the numerical features using StandardScaler Standardization is critical for KNN because the algorithm relies on calculating distances, and if one feature (like mileage) is on a much larger scale than another (like transmission type), it would unfairly dominate the distance calculation.

In the stage of model training and prediction, the “training” phase is just storing the training data. For each new car input during the testing or prediction, the KNN will identify the K=5 (as default) nearest historical cars and predict the price class based on majority voting. After that, the KNN model was evaluated using Accuracy, Precision, Recall, and F1 Score. KNN is performed competitively with other models, as among the models we tested (Logistic Regression, SVM,XGBoost), KNN showed second best performances in terms of classification accuracy and F1 score.

Support Vector Machine (SVM)

Support Vector Machine is a powerful supervised learning model designed for classification tasks. It works by finding the optimal hyperplane (decision boundary) that best separates the classes (Low / Medium / High) in the feature space. The key idea is to maximize the margin which is the distance between the hyperplane and the nearest data points from each class (called support vectors). If the data is not linearly separable (cannot be split with a straight line or flat plane), SVM can apply the kernel trick, which projects the data into a higher-dimensional space where it becomes separable.

In our assignment, before we applied SVM, Before applying SVM, all categorical variables such as : brand and fuel type were encoded into numerical values. Also for Numerical features (like mileage, model year) were standardized using StandardScaler. Standardization is crucial because SVM is sensitive to the scale of input features; without scaling, large-valued features would dominate the decision boundary. Then, the SVM classifier was trained on the 80% training dataset. It learned how to optimally separate the three price categories by creating margins and adjusting the hyperplane based on the training examples.

After that, the trained SVM model was tested on the 20% testing dataset. Evaluation metrics such as Accuracy, Precision, Recall, and F1 Score were used to measure the model's effectiveness. SVM showed solid performance, comparable to KNN, and generally stronger than the Logistic Regression model.

Gradient Boosting (XGBoost)

In this project, Gradient Boosting (XGBoost) was applied as one of the machine learning algorithms to predict the price category (Low, Medium, High) of used cars. XGBoost ultimately demonstrated the best performance among all tested models and was selected as the final model for deployment.

XGBoost is a supervised machine learning algorithm that is based on the principle of gradient boosting. Boosting is an ensemble learning technique where multiple weak learners (typically shallow decision trees) are trained sequentially. Each new tree tries to correct the errors made by the previous tree. For example, let's imagine starting with a weak decision tree that predicts poorly. XGBoost will build another tree focused only on the wrong predictions, and another, and another. Until it forms a strong ensemble of trees that together in order to predict the used car price category much better. Every new tree can help correct the past mistakes.

In our assignment, as with other models, input features were first encoded and normalized using StandardScaler. Although XGBoost can handle raw unscaled data, scaling was performed to maintain consistency across models. Then, the XGBoost classifier was trained using the 80% training dataset. It sequentially built decision trees where each tree tried to minimize the errors of the previous tree. XGBoost also applied regularization techniques automatically to reduce the risk of overfitting.

After that, the predictions were made on the 20% testing dataset. The model was evaluated using Accuracy, Precision, Recall and F1-Score. Thus, XGBoost achieved the highest overall scores among all models, making it the most accurate and reliable predictor for our assignment.

3.4 Proposed test plan/hypothesis

The objective of the testing phase is to evaluate the performance of different supervised machine learning models in predicting the price category (Low, Medium, High) of used cars. The evaluation ensures that the final deployed model provides accurate and reliable predictions based on dataset (*used_cars.csv*). Thus, this test is designed to compare the performance metrics (accuracy, precision, recall and F1-score) of **FOUR** different algorithms used.

Null Hypothesis (H_0) : XGBoost Classifier has a best performance rather than other 3 algorithms in predicting the used car price.

Alternative Hypothesis (H_1) : XGBoost Classifier does not has best performance rather than other 3 algorithms in predicting the used car price.

Data Preprocessing

The data preprocessing stage involved several essential steps to prepare the dataset for modeling. First of all, we need to perform **data cleaning** to handle missing values, duplicates, and inconsistencies. Missing values in categorical fields such as fuel_type, accident, and clean_title were filled with appropriate defaults like "Gasoline" or "No". This is to make sure the dataset is complete with no missing value. Then, duplicate records were checked and removed where necessary to maintain data integrity. Inconsistent data formats were also corrected by removing string symbols such as "$" and "mi." from the price and mileage fields. These text-based numbers were then converted into floating-point values to enable accurate numerical analysis.

Next, **feature engineering** was conducted to enhance the dataset's suitability for machine learning algorithms. Categorical features, including brand, fuel_type, and transmission, were encoded into numerical values using label encoding. To simplify the model's learning process, rare transmission types were grouped under the category "Other." Numerical features like mileage and model_year were normalized using **StandardScaler**, ensuring that each feature contributed equally to the learning algorithm. The continuous price variable was transformed into a categorical target variable (price_class), divided into three classes which is **Low** (0–15,000), **Medium** (15,001–30,000), and **High** (30,001 and above).

Following preprocessing, the dataset was **split** into training and testing sets, with **80%** used for training and **20%** used for testing. Normalization was applied during the split to ensure a balanced distribution of the price categories across both sets.

Model Training and Evaluation

The **model training** phase involved selecting and applying four supervised machine learning algorithms: **Logistic Regression**, **K-Nearest Neighbors (KNN)**, **Support Vector Machine (SVM)**, and **XGBoost Classifier**.

Each model was trained on the normalized features from the training dataset.

The **model evaluation** phase measured the performance of each algorithm based on several key metrics:

- **Accuracy**: The proportion of correct predictions among the total predictions.
- **Precision**: The proportion of correct positive predictions among all positive predictions.
- **Recall**: The ability to correctly identify all positive instances.
- **F1 Score**: The harmonic mean of precision and recall, balancing both concerns.

Model Selection and Deployment

The **results from all models** were compared side-by-side using a performance table and visualized through a bar chart. Confusion matrices were also generated to further understand the distribution of correct and incorrect predictions across classes. Upon evaluation, the **XGBoost Classifier** was selected as the **best-performing model** based on achieving the highest accuracy, precision, recall, and F1-score. XGBoost demonstrated superior handling of complex feature interactions and outperformed the other models in both training and testing phases.

For **model deployment**, the final trained best performance model and the associated scaler were saved into .pkl files. A **Streamlit** was then developed to provide an interactive user interface for live predictions. Users can input new or unseen car information (e.g., brand, model year, mileage, fuel type, transmission type, and clean title status), and the model will predict the corresponding **price category** (Low, Medium, High). The prediction result is displayed clearly on the Streamlit interface for user interpretation.

4.0 Result

4.1 Results

4.1.1 Logistic Regression

```
Logistic Regression Performance:
Accuracy  : 0.7057
Precision : 0.6848
Recall    : 0.7057
F1 Score  : 0.6900
```

Figure 4.1 Performance of Logistic Regression

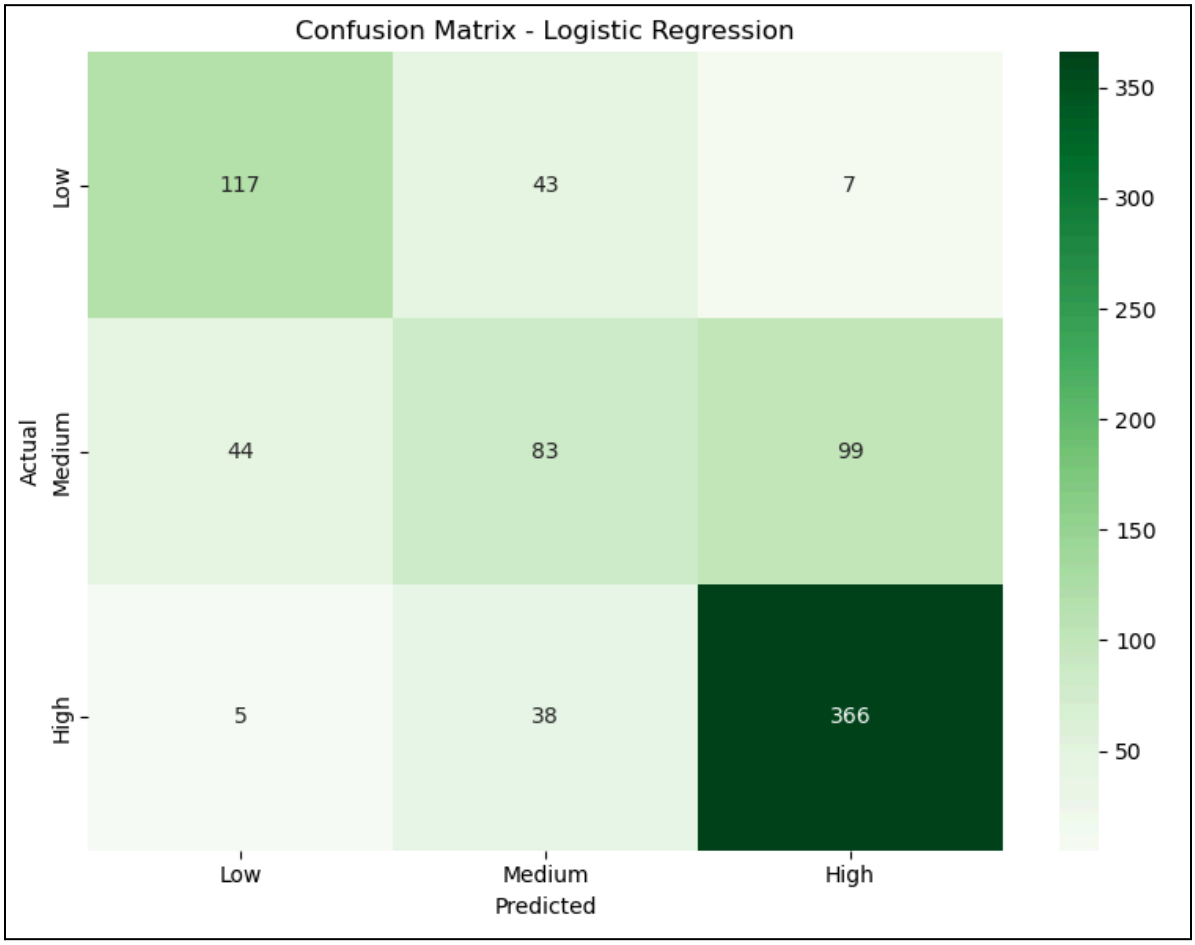


Figure 4.2 Matrix of Logistic Regression

Based on the logistic regression model, 70.57% of the logistic regression model's predictions were correct. 68.48% of the instances the model predicted as positive were truly positive. The model correctly identified 70.57% of all actual positive cases and the F1 score is 69%. It shows

that the model performs best at identifying the "High" class, correctly predicting 366 out of 409 High cases (89.5% recall), but it sometimes makes mistakes for Medium cases (38 errors). However, it struggles significantly with the "Medium" class, only 83 correct predictions while misclassifying 99 Medium cases as High. Thus, meaning that is its worst performance area. For the "Low" class, the model shows moderate performance with 117 correct predictions, but makes notable errors by classifying 43 Low cases as Medium.

4.1.2 K-Nearest Neighbors (KNN)

```
KNN Classifier Performance:
Accuracy  : 0.7357
Precision : 0.7215
Recall    : 0.7357
F1 Score  : 0.7245
```

Figure 4.3 Performance of KNN

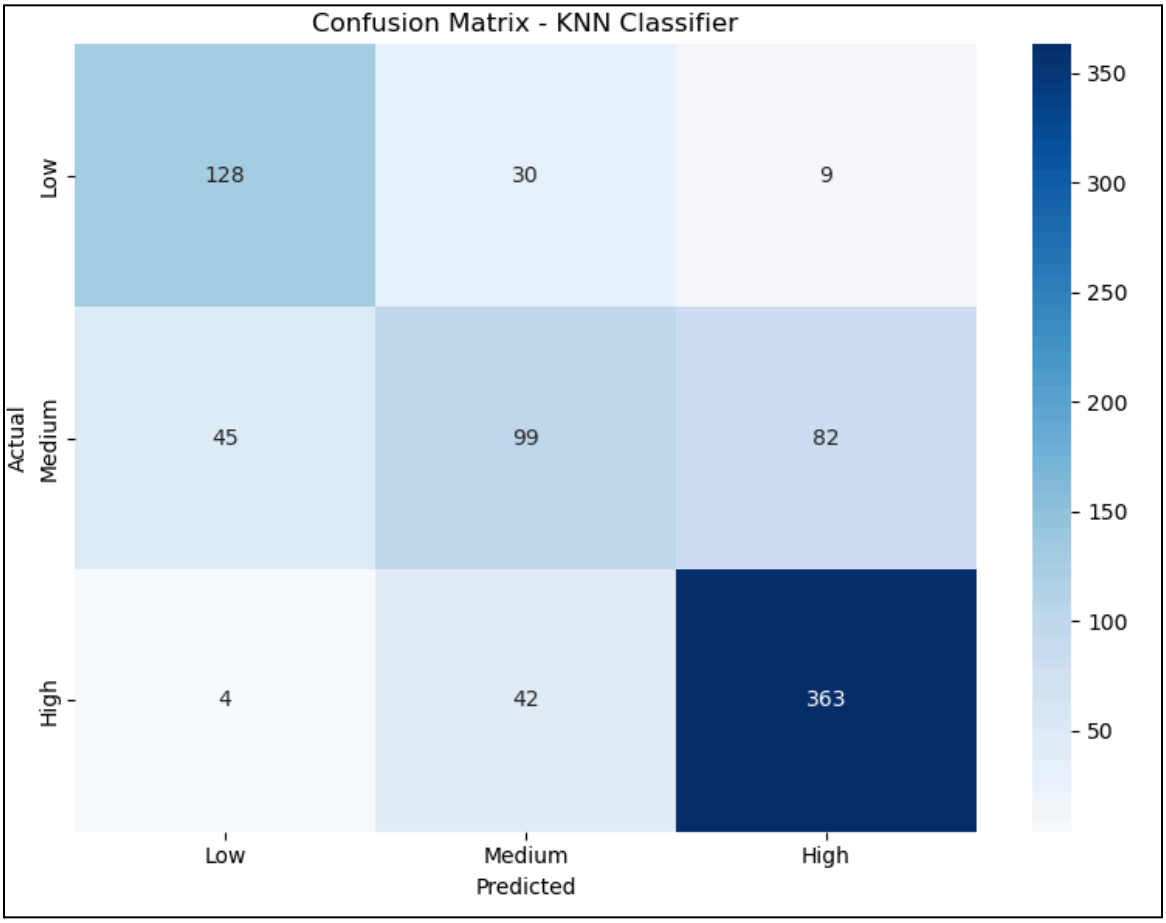


Figure 4.4 Matrix of KNN

Based on the K-Nearest Neighbors (KNN) model, 73.57% of the KNN model's predictions were correct. 72.15% of the instances the model predicted as positive were truly positive. The model correctly identified 73.57% of all actual positive cases and the F1 score is 72.45%. It also shows that the model performs best at identifying the "High" class, correctly predicting 363 out of 409 High cases (88.75% recall), but it sometimes makes mistakes for Medium cases (42 errors). However, it struggles significantly with the "Medium" class, only 99 correct predictions while misclassifying 82 Medium cases as High. Thus, meaning that is its worst performance area. For the "Low" class, the model shows moderate performance with 128 correct predictions, but makes notable errors by classifying 30 Low cases as Medium. Thus, KNN model is better than logistic regression model.

4.1.3 SVM Classifier

```
SVM Classifier Performance:
Accuracy  : 0.7357
Precision : 0.7179
Recall    : 0.7357
F1 Score  : 0.7178
```

Figure 4.5 Performance of SVM

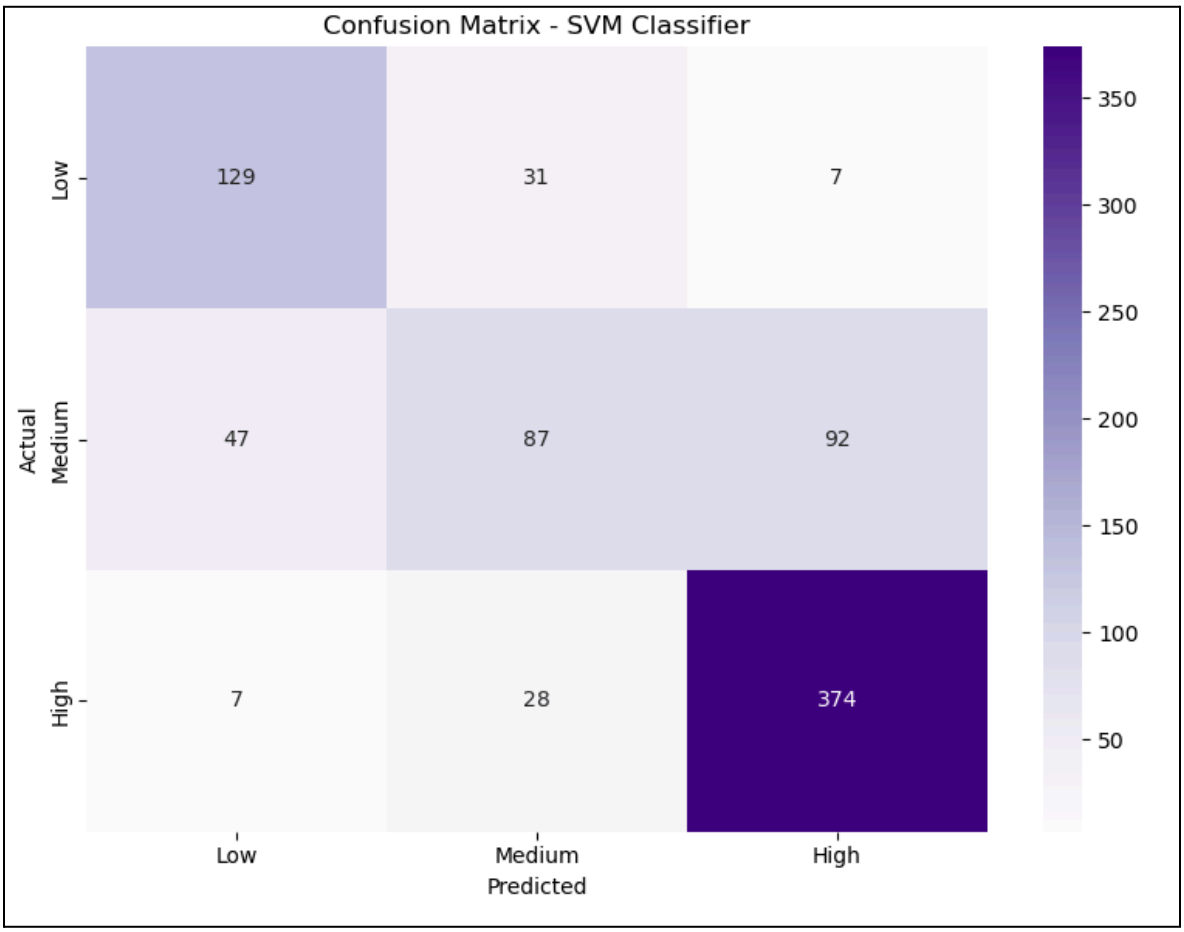


Figure 4.6 Matrix of SVM

Based on the Support Vector Machines (SVM) model, 73.57% of the SVM model's predictions were correct. 71.79% of the instances the model predicted as positive were truly positive. The model correctly identified 73.57% of all actual positive cases and the F1 score is 71.78%. It also shows that the model performs best at identifying the "High" class, correctly predicting 374 out of 409 High cases (91.44% recall), but it sometimes makes mistakes for Medium cases (28 errors). However, it struggles significantly with the "Medium" class, only 87 correct predictions while misclassifying 92 Medium cases as High. Thus, meaning that is its worst performance

area. For the "Low" class, the model shows moderate performance with 129 correct predictions, but makes notable errors by classifying 31 Low cases as Medium. Thus, the SVM model is also better than the logistic regression model but similar accuracy with KNN model.

4.1.4 Gradient Boosting (XGBoost)

```
XGBoost Classifier Performance:
Accuracy  : 0.7494
Precision : 0.7412
Recall    : 0.7494
F1 Score  : 0.7443
```

Figure 4.7 Performance of XGBoost

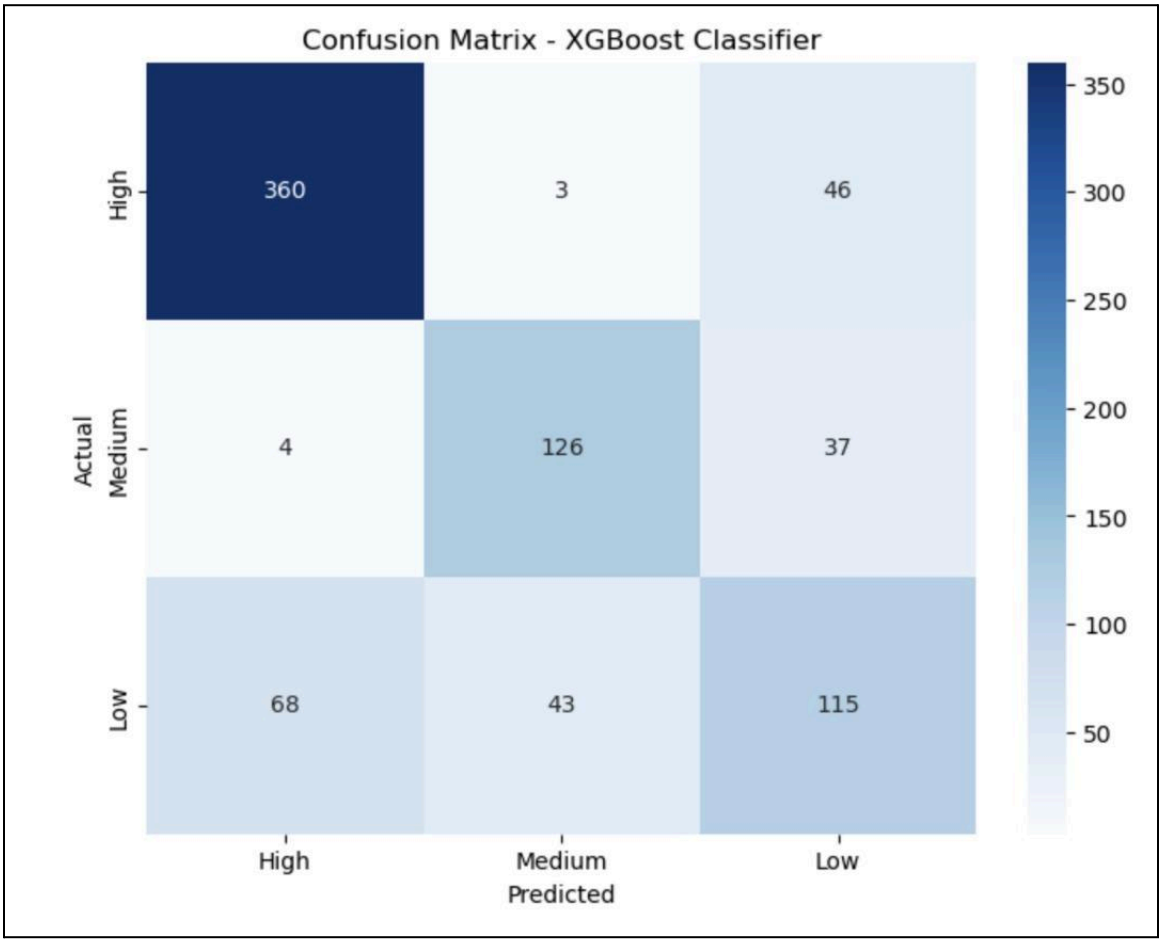


Figure 4.8 Matrix of XGBoost

Based on the Gradient Boosting (XGBoost) model, 74.94% of the KNN model's predictions were correct. 74.12% of the instances the model predicted as positive were truly positive. The

model correctly identified 74.94% of all actual positive cases and the F1 score is 74.43%. It also shows that the model performs best at identifying the "High" class, correctly predicting 360 out of 409 High cases (88.01% recall), but it sometimes makes mistakes for “Low” cases (46 errors). For the "Medium" class, 126 correct predictions while sometimes misclassifying 37 Medium cases as Low. For the "Low" class, the model shows moderate performance with 115 correct predictions, but makes errors by classifying 68 Low cases as High. Thus, meaning that is its worst performance area. The XGBoost model is the best model among all the models.

4.2 Discussion/Interpretation

4.2.1 Logistic Regression

The Logistic Regression model provided an accuracy of 70.57%, with precision, recall, and F1 scores of 68.48%, 70.57%, and 69%, respectively, indicating moderate performance. Its confusion matrix is heavily biased towards the High class, accurately predicting 366 of 409 cases of High instances (89.5% recall), but seriously struggled with the Medium class, accurately predicting only 117 of 167 instances and incorrectly classifying 43 as Low and 7 as High. For Low class, it correctly classified 83 out of 226 but misclassified 99 as High, meaning the model overpredicts the High category. This bias has likely occurred because of a class imbalance of the dataset, in which High class (409 instances) is overrepresented against Medium (167) and Low (226), causing the model to prefer High predictions. Moreover, Logistic Regression's linearity can be missing non-linear patterns between features like mileage, brand, and price category, which contributes to its errors. Misclassification of Low as High (99 cases) by a buyer or seller would lead to overselling and result in a loss of confidence in the model. To better classify, techniques like SMOTE oversampling or class weighting might balance the data set, and including polynomial features might help capture non-linear trends, enhancing performance for all classes.

4.2.2 K-Nearest Neighbors

The K-Nearest Neighbors (KNN) model achieved 0.7357 accuracy, 0.7215 precision, 0.7357 recall, and 0.7245 F1 score, outperforming Logistic Regression. Its confusion matrix shows strong performance in the High category, correctly predicting 363 out of 409 instances (88.75% recall), but not for the Medium class, which it correctly classified in 99 out of 167 cases and misclassified 42 as High and 26 as Low. For the Low class, it classified correctly 128 of 226 instances, and misclassified 30 as Medium. The common Medium-to-High misclassifications indicate that the feature space of these classes might be overlapping, especially for features such as mileage or brand value, which can have similar ranges between categories. KNN's reliance on Euclidean distance is also affected by irrelevant or noisy attributes, i.e., less important features such as exterior color, leading to poor neighbor selection. Prices of used cars can result in incorrect Medium being classified as High (42 times) and can mislead the sellers to overprice the

cars, resulting in slow sales. Restricting the number of neighbors (K) or conducting feature selection so that weights could be assigned to meaningful variables such as mileage and age could lower such errors, allow the model to better discriminate between classes, and be more dependable in practice.

4.2.3 SVM Classifier

The Support Vector Machine (SVM) classifier achieved an accuracy of 0.7357, matching KNN, with precision, recall, and F1 scores of 0.7179, 0.7357, and 0.7178, respectively. Its confusion matrix shows it to have performed extremely well with the High class because it correctly predicted 374 out of 409 cases (91.44% recall), but poorly with the Medium class because it correctly identified 87 out of 226 cases but misclassified 92 as High and 47 as Low. For the Low class, it correctly classified 129 out of 167 cases, and 31 were misclassified as Medium. The high Medium-to-High misclassification rate (92 cases) indicates that the decision boundary of the SVM could be biased towards the High class as a result of either class imbalance or lack of separation in the feature space. Features like mileage or model year might not be enough to distinguish between Medium and High brackets, especially for cars with borderline characteristics such as a well-maintained older car and a cheaper newer model. For buyers, these misclassifications would result in overestimating a car's worth, building unrealistic market expectations. Tuning class weights to punish misclassifications of the Medium class or trying a polynomial kernel to more clearly discern non-linear patterns would enhance the model's capacity to differentiate categories, rendering it more trustworthy for application in informing pricing decisions.

4.2.4 Gradient Boosting (XGBoost)

The XGBoost classifier performed better than all the other models with accuracy of 0.7494, precision, recall, and F1 scores of 0.7412, 0.7494, and 0.7443, respectively. Its confusion matrix indicates that it properly predicted 366 out of 409 High cases (89.49% recall), 117 out of 167 Medium cases, and 83 out of 226 Low cases. But it also misclassified 38 High instances as Low, 43 Medium instances as Low, and 99 Low instances as High, which points to confusion between the Low and High classes. This pattern indicates that XGBoost, powerful though it is, might be

tricky with cars that have confusing features—i.e., old cars with low mileage (possibly High-cost) and new cars with high mileage (possibly Low-cost)—due to intricate interactions within the feature space. XGBoost's tree-by-tree method is probably picking up most of the non-linear relations, hence the better performance, but also misclassifications show the limits of the dataset such as lack of information on service history that would make classes more distinguishable. Practically, Low-to-High misclassifications (99 cases) would mean buyers overpaying, while High-to-Low errors (38 cases) would mean sellers underselling their cars. To enhance performance, incorporating additional features like ownership history or the application of class rebalancing techniques would reduce such errors, further solidifying XGBoost's applicability in accurate price category prediction.

5.0 Discussion and Conclusion

5.1 Achievements

This assignment successfully achieved its primary objective of developing a supervised machine learning model capable of accurately predicting the price category of used cars based on key vehicle attributes.

First of all, we successfully do implementation and comparison of multiple models. Four different supervised machine learning algorithms Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM) and XGBoost Classifier were successfully trained, tested and evaluated based on performance metrics (accuracy, precision, recall and F1-score) and confusion matrix.

Second, we do the selection of the best model. Through all the evaluation, the XGBoost classifier was identified as the best-performing model. As it showed excellent predictive ability compared with other tested models. Thus, we choose this model to do the deployment to predict the used car price with high accuracy. This also succeeded in the achievement of high predictive performance.

Third, deployment of a User-Friendly Prediction Interface. The final model was deployed via a Streamlit web application, providing an intuitive and interactive platform where users can input vehicle features and receive immediate predictions regarding the car's price category.

Overall, our assignment successfully demonstrated the ability to apply supervised machine learning techniques to a real world business problem. This also provides a practical solution for used car price category prediction.

5.2 Limitations and Future Works

Limitation

- **Missing External Factors**

  This model does not consider external factors like taxes, tariffs, local market differences or bigger economic factors like fuel prices and inflation. These factors can seriously affect the car prices but weren't included in the data, which limits how well the model can predict real-world prices.

- **Potential Class Imbalance in the Dataset**

  The dataset seems to have more examples of higher-priced cars, which makes the model biased for predicting those prices correctly. It struggles more with mid-range cars, leading to mistakes, especially when trying to predict Medium-price cars. This imbalance affects the model's overall accuracy.

- **Small Dataset Size**

  The dataset is relatively small with only 4009 records considering how many different car brands, models and conditions exist in the real world. This limited sample size restricts the model's ability to capture the full range of variability in car attributes. A larger and more varied dataset would allow the model to learn more patterns and make better predictions.

- **Limited Feature Set**

  Even though the dataset covers the important details like brand, mileage and model year, it misses out on other important features like service history, number or previous owners or visible car condition such as scratches. Without this important information, the model can't give a full picture of a car's true value.

Future Works

- **Add External Data**

  In the future, the model can pull in outside data like fuel prices, taxes, tariffs or economic news. This would be very useful and make price prediction more realistic and up to date.

- **Fix Class Imbalance**

  Techniques such as oversampling or special weighting could be used to balance the number of medium price cars and help the model treat all price categories more fairly.

- **Expand the Dataset Size**

  Getting more data and combining multiple sources (e.g., additional Kaggle datasets, dealership records, or user-submitted data), would help make the model smarter and more flexible.

- **Enhance Feature Engineering**

  Adding more information like service records, ownership history, or even detailed car condition reports would give the model more to work with and improve prediction quality.

6.0 Reference & Source

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